

creation of a model or virtual platform in the electronics domain. Once you have a virtual platform, you can share the same virtual system with anyone you want, and eliminate hardware availability issues; meaning that you kill the catch-22 classic situation of not having hardware slow-down software development."

There have been many recent advances in design and simulation tools that have made the whole process of resource optimisation even more effective. A few are enumerated below:

Runtime enhancements. Simulation tools have enhancements in terms of runtime so as to get faster results. They also support co-simulation on real hardware to further speed-up the lengthy and complex simulations by providing hardware acceleration. Simulation tools now also come with seamless interfaces to ECAD tools, which help users import layout data and create spatial layered geometries automatically.

Advances in design tools. "The design map tools these days offer new features like 'large block placer,' 'register ordering' and 'routing congestion aware algorithms' to optimise the resource-usage. The tools are even capable of gating clocks and disabling embedded memories to reduce the power consumption," informs Jain. Further to this, the tools help engineers time-multiple various algorithms through features like 'partial reconfiguration'. This saves tremendous amount of resources and power.

Virtual prototyping. This is another innovative methodology of combining mechanical modelling and simulation with control design to increase the efficiency of designing and prototyping. "With virtual prototyping, you can connect your software design and control algorithms to your 3D CAD mechanical models to test the mechanics of your system before even building your first physical prototype. "Virtual prototyping lowers the risks associated with machine design by improving the understanding of cus-

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tomers requirements, speeding up the design process, and streamlining debugging. Without a virtual prototype, you would have to build the entire physical prototype before you could obtain tangible customer-feedback on the product's operation. By using virtual prototyping, you can show a digital representation of the mechanics of the machine to the customer and get feedback more quickly and easily before actually building the machine," explains Mohanram. This ensures that customers are more involved in the

design process and prevents you from having to wait until it is too late in the prototyping process to get customer feedback.

Multiphysics modelling. This is a simulation process of accurately capturing real-life processes to help speed up the product design process by minimising the prototype phase. This helps to reduce resource consumption right from ideation to product manufacture. Modelling helps visualise new concepts and ideas inexpensively and quickly. As you move further along, the behaviour of various components interacting with each other can also be simulated. Manufacturing processes can also be optimised using simulation tools. Dr Dravid illustrates, "Real-life processes must often involve the interaction of several physical phenomena in tandem. An example would be predicting heating in electronic chips. This process involves the phenomenon of current flowing through the chip, coupled with the phenomenon of heat generation due to current flow. This is an example of a multiphysics problem. Thus, multiphysics modelling is essential to accurately capture real life processes, which involve several such phenomena."

In a nutshell

Considering the crucial role that resource optimisation can play in the lifecycle of an embedded system or device, and hazards that neglecting this key engineering principle can lead to, it is only worthwhile that design engineers exercise caution and adopt smart practices and tools to ensure that they never lose sight and control over resources that they have in their repertoire—for this is indeed a core requirement for achieving meaningful efficiencies in the design and development of embedded devices and chips. ●

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